

1. General Description

The EM74HC574 and EM74HCT574 are 8-bit D-type transparent latches with 3-state outputs. The device features a clock (CP) and output enable ($\overline{OE}$) inputs. The flip-flops will store the state of their individual D-inputs that meet the set-up and hold time requirements on the LOW-to-HIGH clock (CP) transition. A HIGH on $\overline{OE}$ causes the outputs to assume a high-impedance OFF-state. Operation of the $\overline{OE}$ input does not affect the state of the flip-flops. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

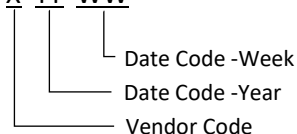
- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Input levels:
 - For EM74HC574: CMOS level
 - For EM74HCT574: TTL level
- Inputs and outputs on opposite sides of package allowing easy interface with microprocessors
- Useful as input or output port for microprocessors and microcomputers
- 3-state non-inverting outputs for bus-oriented applications
- Common 3-state output enable input
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

Octal D-type flip-flop; positive edge-trigger; 3-state

Table 1. Ordering information

MARKING INFORMATION

X YY WW



Block diagram of the 74F175 8-bit D-type flip-flop with 3-state outputs. The diagram shows two main blocks: "FF1 to FF8" and "3-STATE OUTPUTS".

- FF1 to FF8:** This block has 8 data inputs (D0 through D7) on the left and 8 data outputs (Q0 through Q7) on the right. It is labeled "FF1 to FF8".
- 3-STATE OUTPUTS:** This block has 8 data outputs (Q0 through Q7) on the right. It is labeled "3-STATE OUTPUTS".
- Control Inputs:**
 - CP (Clock Pulse):** Input 11, connected to the clock input of both blocks.
 - OE (Output Enable):** Input 1, connected to the output enable input of both blocks.
- Data Inputs/Outputs:**
 - Inputs: D0, D1, D2, D3, D4, D5, D6, D7 (pins 2 through 9).
 - Outputs: Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7 (pins 12 through 19).

Fig. 1. Functional diagram

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Octal D-type flip-flop; positive edge-trigger; 3-state

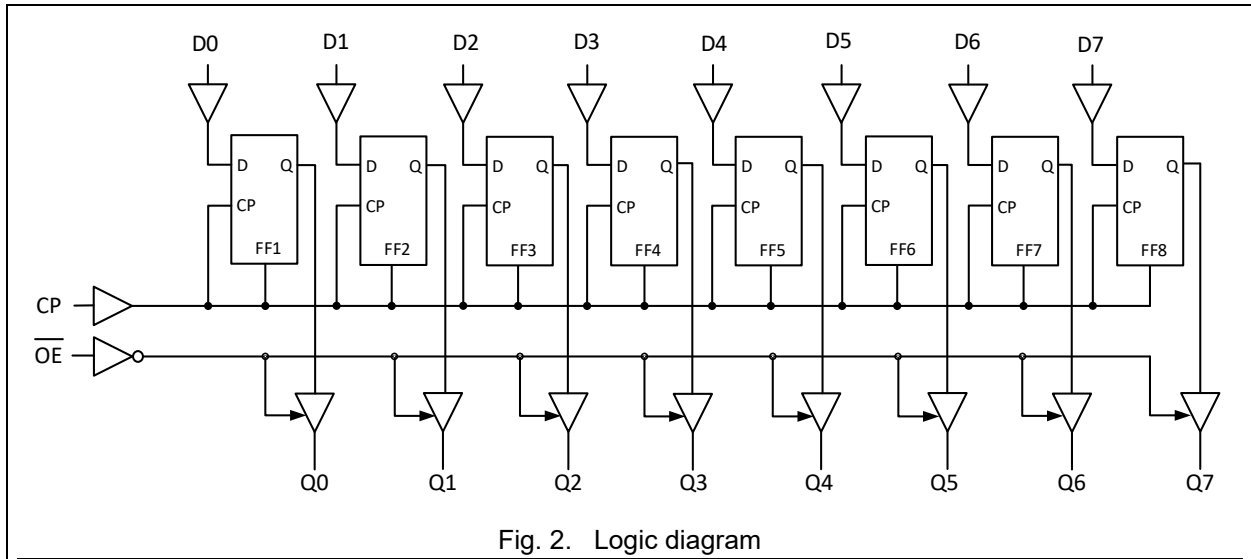


Fig. 2. Logic diagram

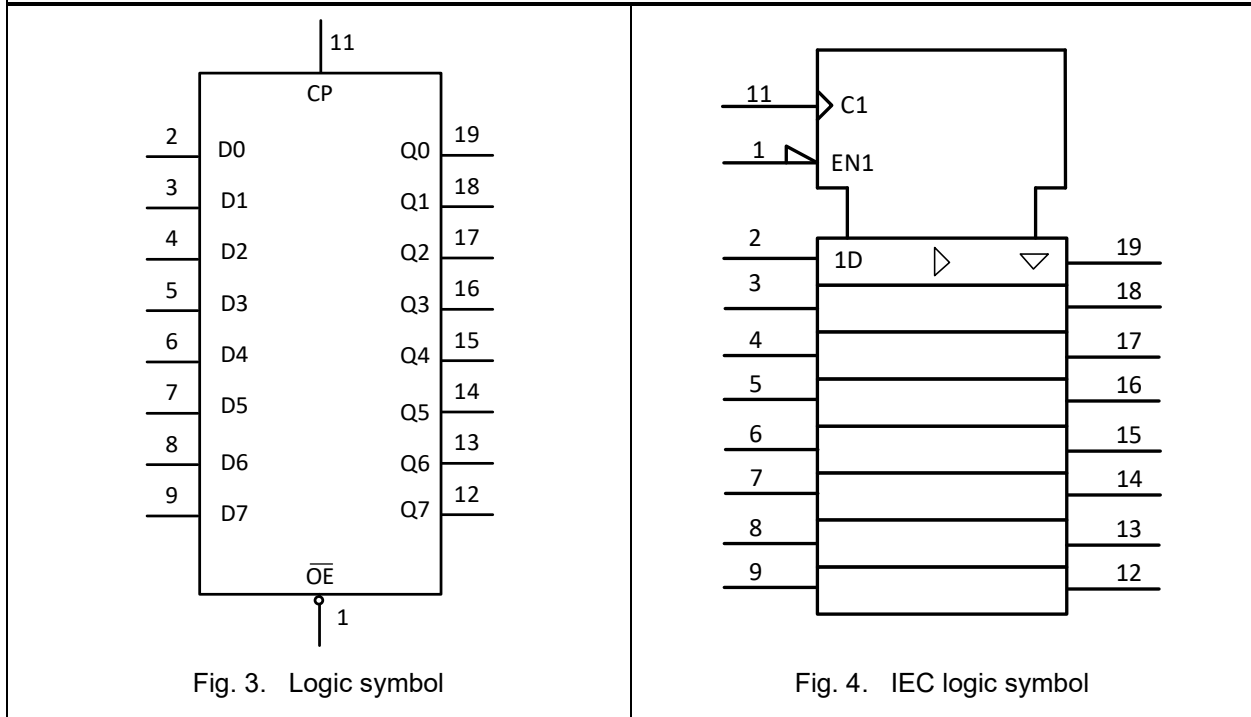


Fig. 3. Logic symbol

Fig. 4. IEC logic symbol

5. Pinning Information

5.1. Pinning

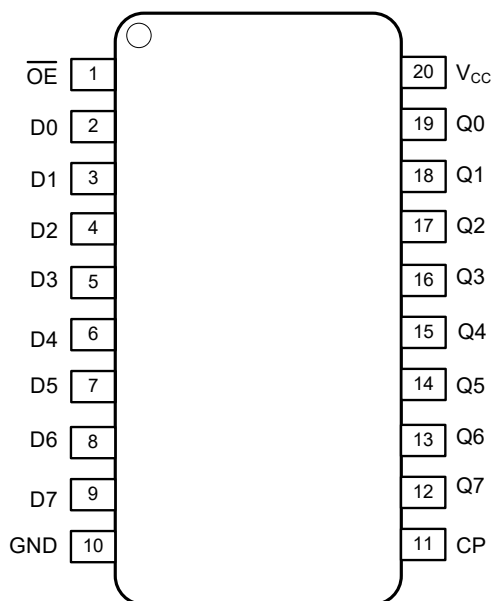


Fig. 5. Top view pin configuration SOP-20L and TSSOP-20L

5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
$\overline{OE}$	1	3-state output enable input (active LOW)
Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7	19,18,17,16,15,14,13,12	3-state flip-flop outputs
D0, D1, D2, D3, D4, D5, D6, D7	2, 3, 4, 5, 6, 7, 8, 9	data inputs
GND	10	ground (0 V)
CP	11	clock input (LOW-to-HIGH, edge triggered)
V _{cc}	20	supply voltage

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6.Functional Description

Table 3. Function table

H = HIGH voltage level; h = HIGH voltage level one set-up time prior to the LOW-to-HIGH clock transition;

L = LOW voltage level; l = LOW voltage level one set-up time prior to the LOW-to-HIGH clock transition;

Z = high-impedance OFF-state; $\uparrow$ = LOW-to-HIGH clock transition.

Operating modes	Input			Internal flip-flop	Outputs
	$\overline{OE}$	CP	Dn		Qn
Load and read register	L	$\uparrow$	l	L	L
	L	$\uparrow$	h	H	H
Load register and disable output	H	$\uparrow$	l	L	Z
	H	$\uparrow$	h	H	Z

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7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 5. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$		± 20	mA
I_O	output current	$-0.5\text{ V} < V_O < V_{CC} + 0.5\text{ V}$		± 35	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation	$T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$		500	mW
T_{stg}	storage temperature		-65	+150	$^{\circ}\text{C}$

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 6. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC574			EM74HCT574			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	+25	+125	-40	+25	+125	$^{\circ}\text{C}$
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.0\text{ V}$			625				ns/V
		$V_{CC} = 4.5\text{ V}$		1.67	139		1.67	139	ns/V
		$V_{CC} = 6.0\text{ V}$			83				ns/V

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9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC574								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 4.5 V	3.15			3.15		V
		V _{CC} = 6.0 V	4.2			4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 4.5 V			1.35		1.35	V
		V _{CC} = 6.0 V			1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20 μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20 μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20 μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -6.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -7.8 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20 μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 6.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 7.8 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 6.0V ; V _O = V _{CC} or GND			±5		±10	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			3.5				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HCT574								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0			2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = -20 μA	4.4		4.4		V	
		I _O = -6.0 mA	3.84		3.7		V	
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = 20 μA			0.1		0.1	V
		I _O = 6.0 mA			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	μA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 5.5 V ; V _O = V _{CC} or GND			±5		±10	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V			20		40	μA
ΔI _{CC}	additional supply current	per pin ; V _I = V _{CC} – 2.1 V ; I _O = 0 A ; other inputs at V _{CC} or GND ;V _{CC} = 4.5 V to 5.5 V			450		490	μA
C _I	input capacitance			3.5				pF

[1]All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

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Octal D-type flip-flop; positive edge-trigger; 3-state

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 9.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC574								
t _{pd}	propagation delay	CP to Qn; see Fig. 6 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		18	ns
t _{en}	enable time	$\overline{OE}$ to Qn; see Fig. 8 [3]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		22	ns
		V _{CC} = 6.0 V			13		16	ns
t _{dis}	disable time	$\overline{OE}$ to Qn; see Fig. 8 [4]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		22	ns
		V _{CC} = 6.0 V			13		16	ns
t _t	transition time	Qn; see Fig. 6 [5]						
		V _{CC} = 2.0 V			25		30	ns
		V _{CC} = 4.5 V			7		9	ns
		V _{CC} = 6.0 V			7		9	ns
t _w	pulse width	CP HIGH or LOW; see Fig. 7						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
t _{su}	set up time	Dn to CP; see Fig.7						
		V _{CC} = 2.0 V	75			90		ns
		V _{CC} = 4.5 V	15			18		ns
		V _{CC} = 6.0 V	13			15		ns
t _h	hold time	Dn to CP; see Fig.7						
		V _{CC} = 2.0 V	5			5		ns
		V _{CC} = 4.5 V	5			5		ns
		V _{CC} = 6.0 V	5			5		ns

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max			
$f_{\max}$	maximum frequency	CP; see Fig. 6						
		$V_{CC} = 2.0 \text{ V}$	4.8			4.0		MHz
		$V_{CC} = 4.5 \text{ V}$	24			20		MHz
		$V_{CC} = 6.0 \text{ V}$	28			24		MHz
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz}$; $V_I = \text{GND to } V_{CC}$; [6]		30				pF
EM74HCT574								
t_{pd}	propagation delay	CP to Qn; see Fig. 6 [2]						
		$V_{CC} = 4.5 \text{ V}$			25		30	ns
t_t	transition time	Qn; see Fig. 6 [5]						
		$V_{CC} = 4.5 \text{ V}$			7		9	ns
t_{en}	enable time	$\overline{OE}$ to Qn; see Fig. 8 [3]						
		$V_{CC} = 4.5 \text{ V}$			20		25	ns
t_{dis}	disable time	$\overline{OE}$ to Qn; see Fig. 8 [4]						
		$V_{CC} = 4.5 \text{ V}$			20		25	ns
t_w	pulse width	CP HIGH or LOW; see Fig. 7						
		$V_{CC} = 4.5 \text{ V}$	20			24		ns
t_{SU}	set up time	Dn to CP; see Fig.7						
		$V_{CC} = 4.5 \text{ V}$	15			18		ns
t_h	hold time	Dn to CP; see Fig.7						
		$V_{CC} = 4.5 \text{ V}$	5			5		ns
$f_{\max}$	maximum frequency	CP; see Fig.6						
		$V_{CC} = 4.5 \text{ V}$	24			20		ns
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz}$; $V_I = \text{GND to } V_{CC} - 1.5 \text{ V}$; [6]		30				pF

[1] Typical values are measured at $T_{\text{amb}} = 25 \text{ }^{\circ}\text{C}$.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] t_{en} is the same as t_{PZL} and t_{PZH} .

[4] t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[5] t_t is the same as t_{THL} and t_{TLH} .

[6] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

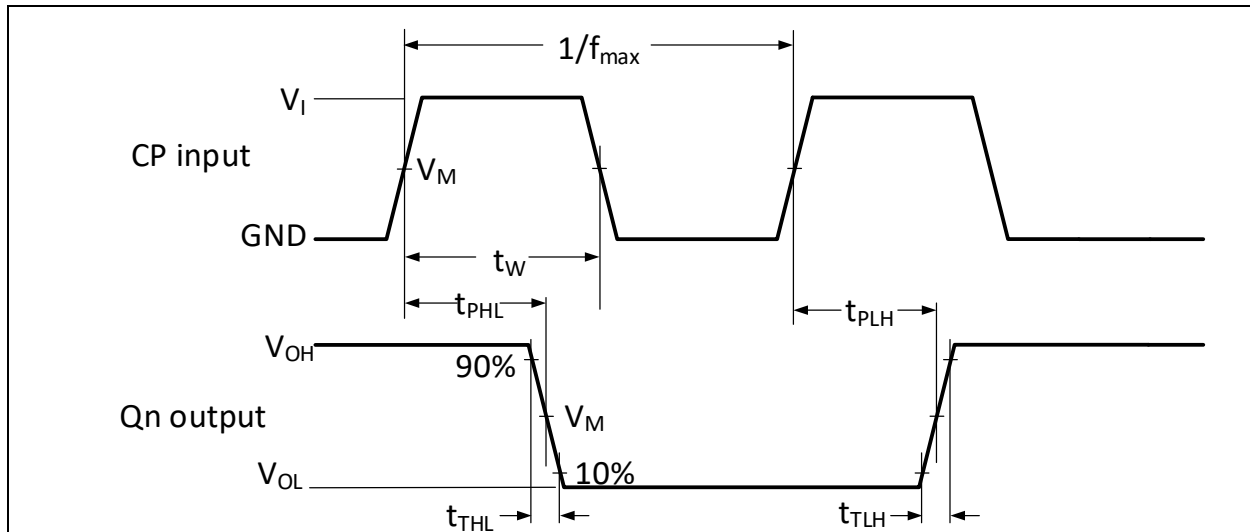
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\sum (C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

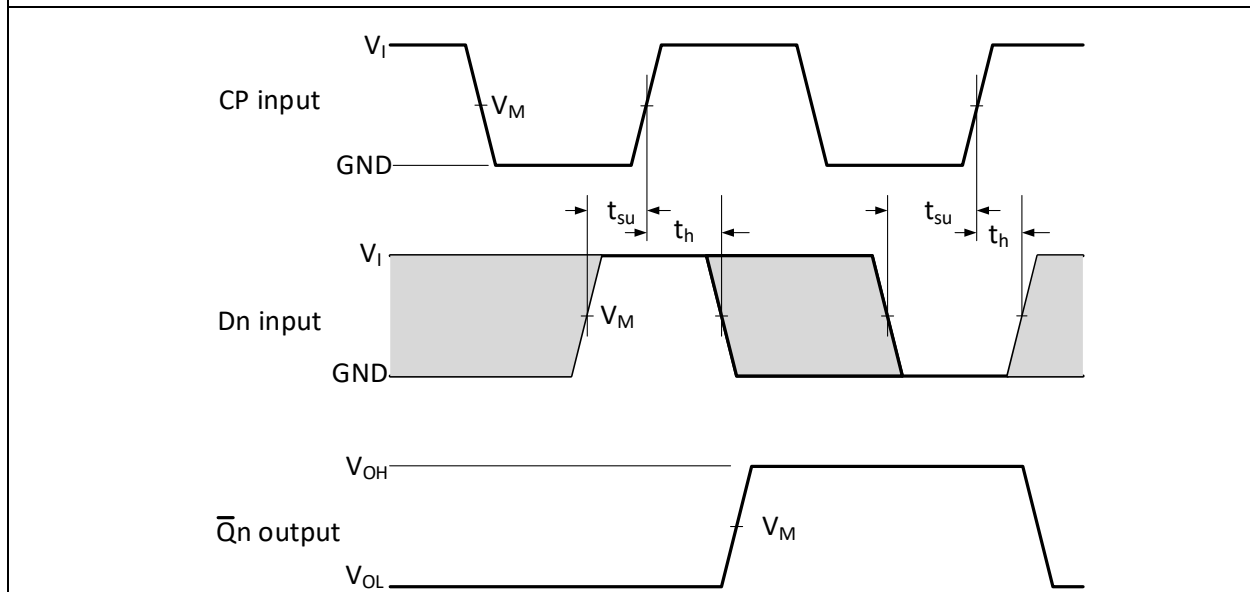
10.1. Waveforms and test circuit



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

Fig. 6. Propagation delay input (CP) to output (Qn), output transition time, clock input (CP) pulse width and the maximum frequency (CP)



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 7. The data input (D) to clock input (CP) set-up times and clock input (CP) to data input (D) hold times

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Octal D-type flip-flop; positive edge-trigger; 3-state

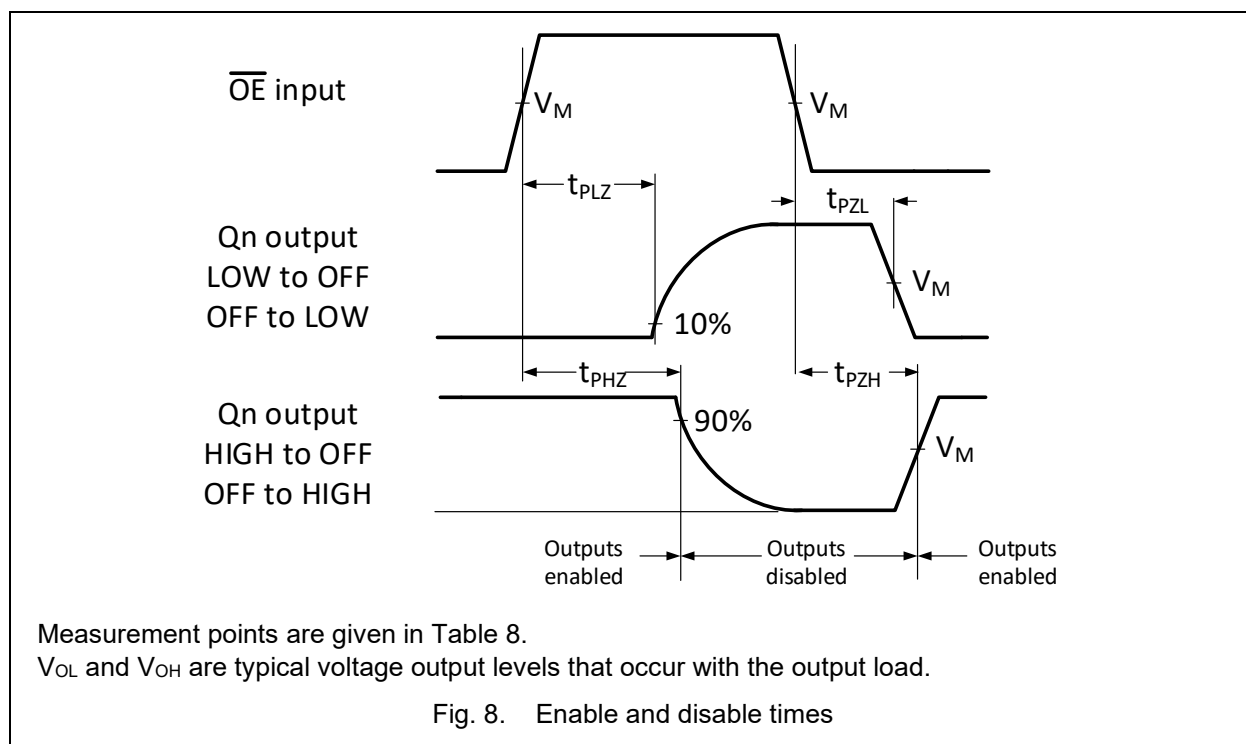


Table 8. Measurement points

Type	Input	Output		
	V_M	V_M	V_X	V_Y
EM74HC574	$0.5V_{CC}$	$0.5V_{CC}$	$0.1V_{CC}$	$0.9V_{CC}$
EM74HCT574	1.3 V	1.3 V	$0.1V_{CC}$	$0.9V_{CC}$

EM74HC574; EM74HCT574

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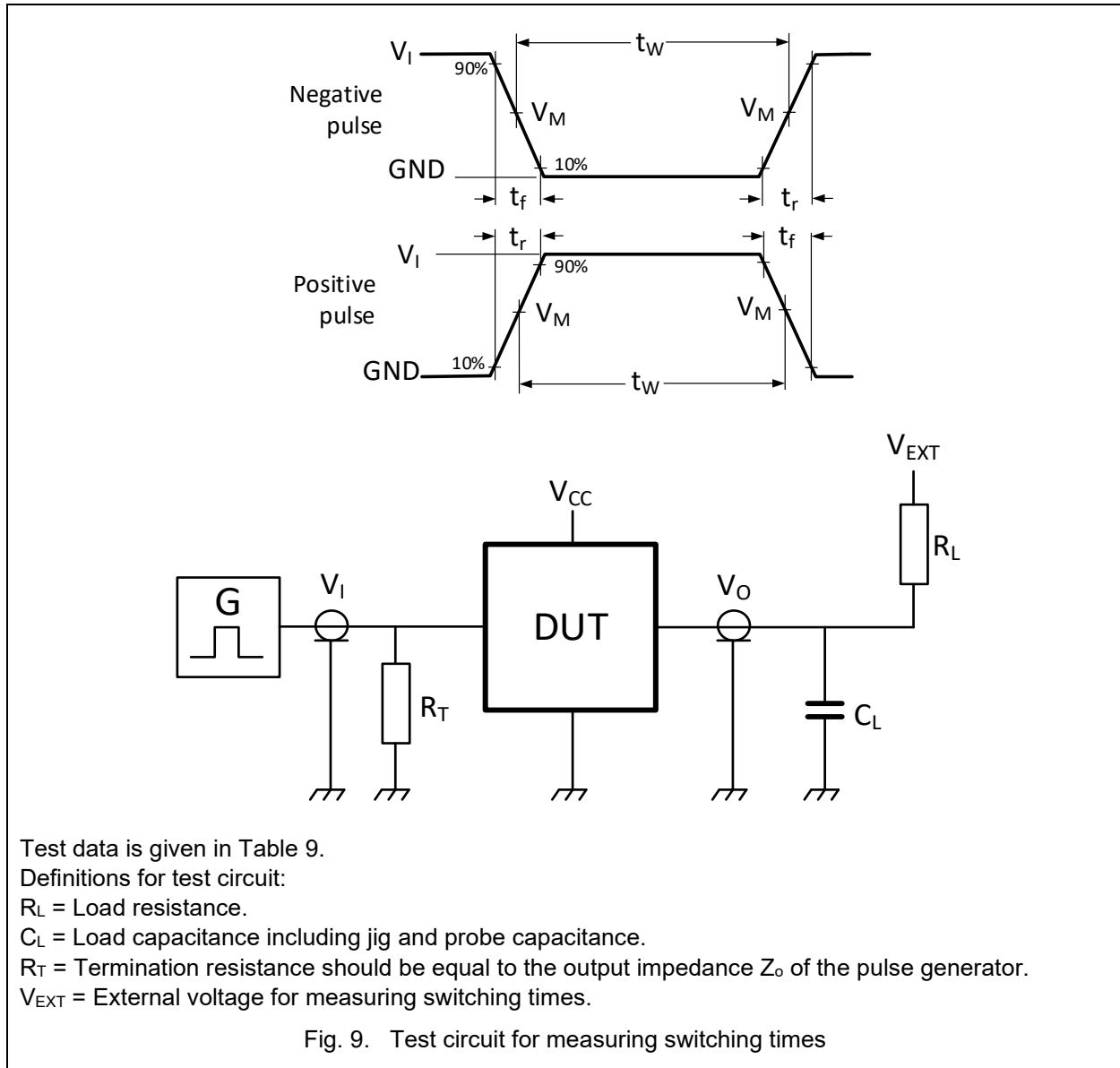
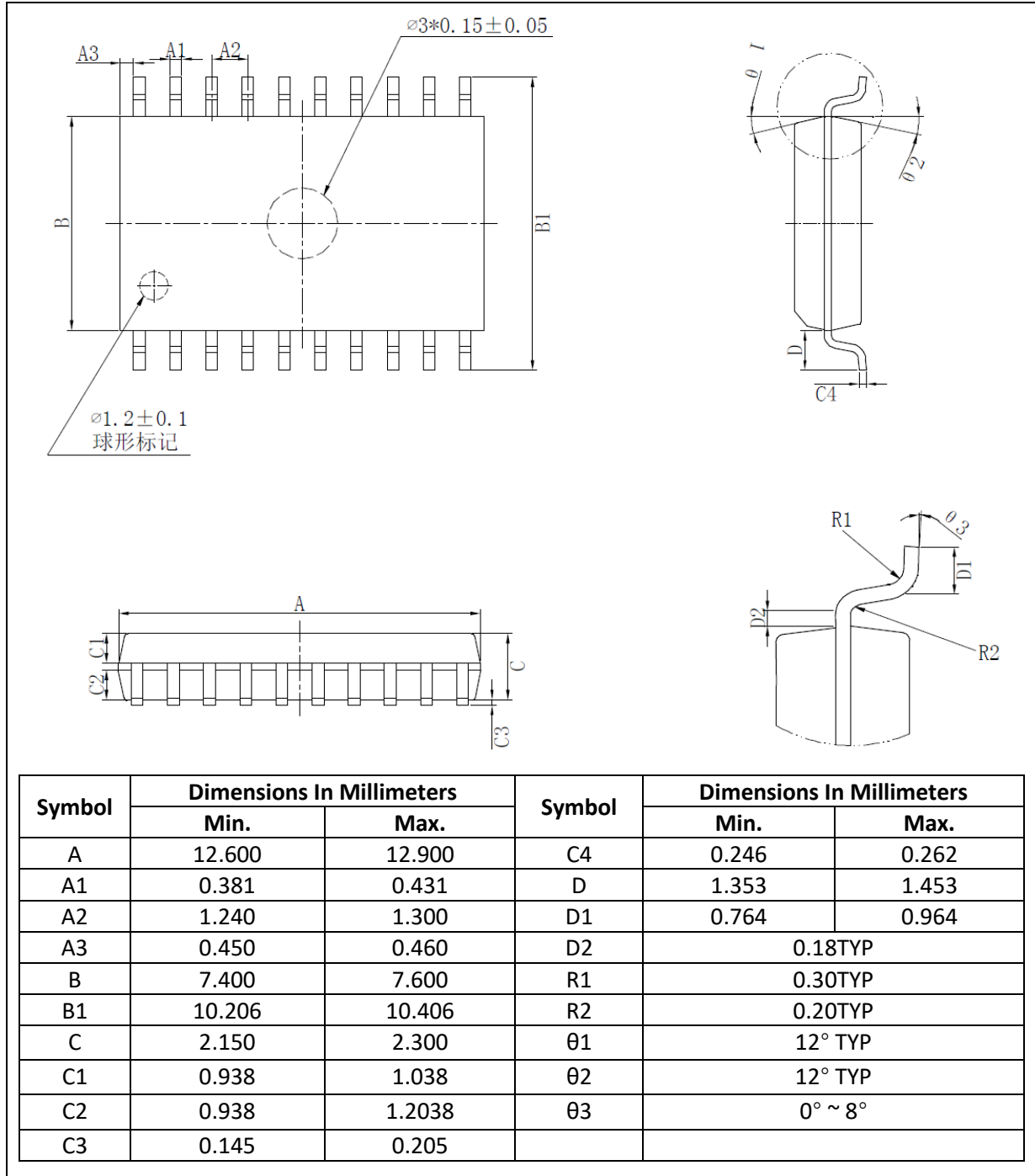


Table 9. Test data

Type	Input		Load		V_{EXT}		
	V_I	$t_r = t_f$	C_L	R_L	t_{PZL}, t_{PLZ}	t_{PHL}, t_{PLH}	t_{PZH}, t_{PHZ}
EM74HC574	V_{CC}	2.5 ns	15 pF	500 Ω	V_{CC}	Open	GND
EM74HCT574	3 V	2.5 ns	15 pF	500 Ω	V_{CC}	Open	GND

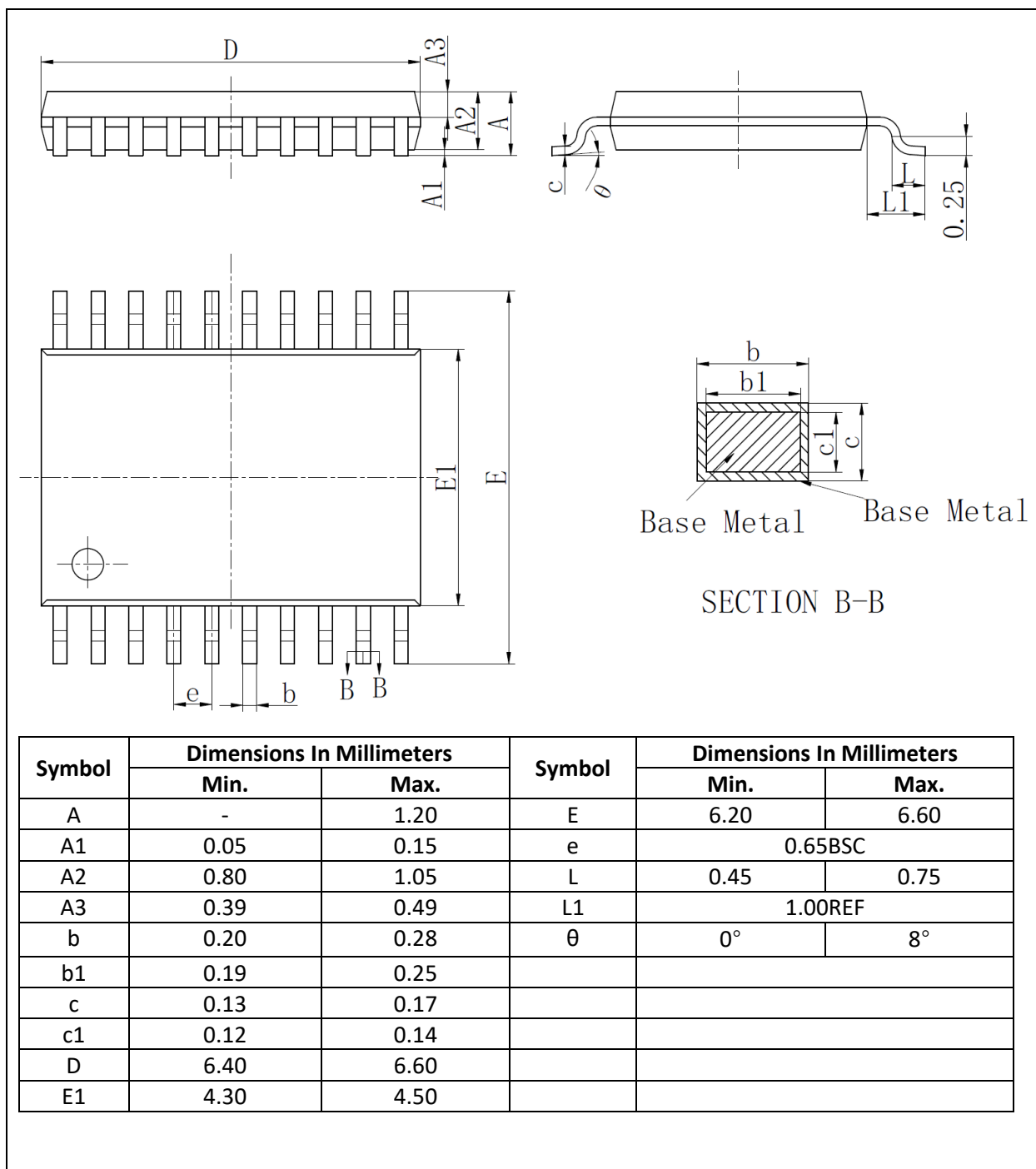
11. Package Outline

SOP-20L


EM74HC574; EM74HCT574

Octal D-type flip-flop; positive edge-trigger; 3-state

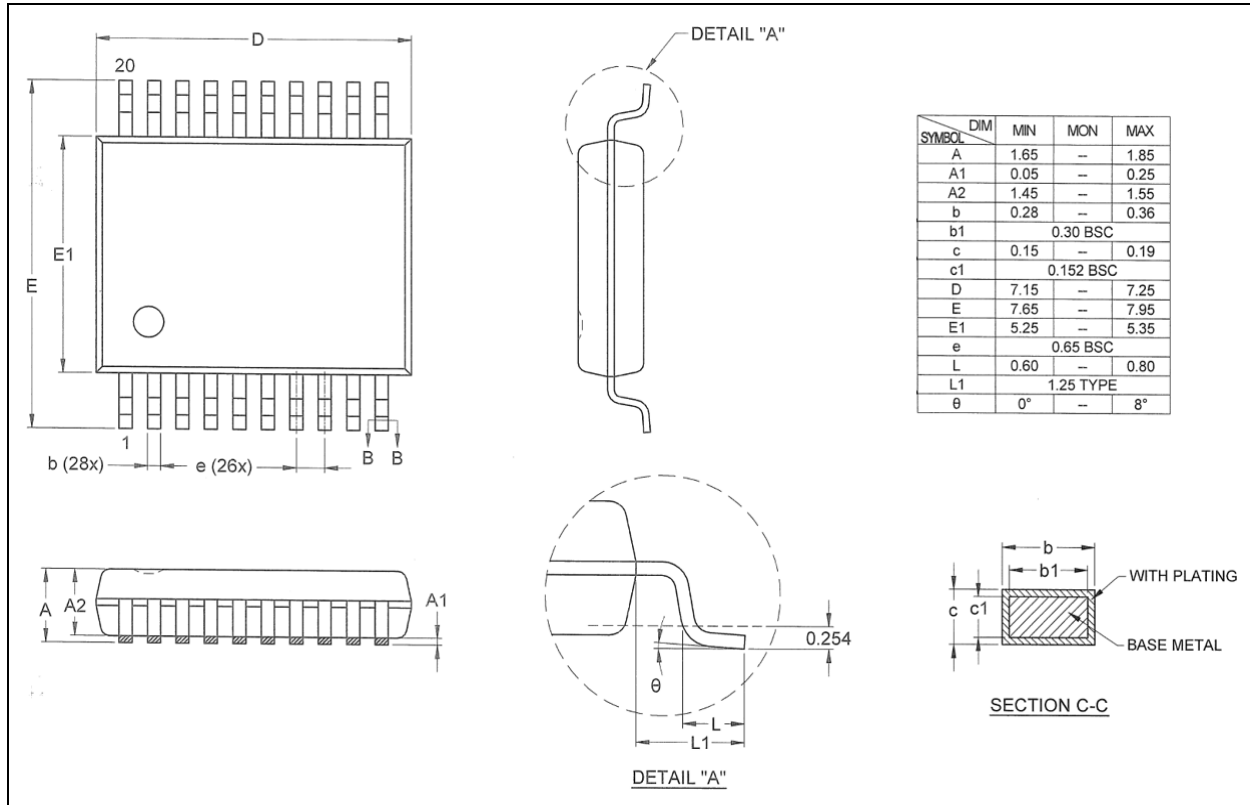
TSSOP-20L



EM74HC574; EM74HCT574

Octal D-type flip-flop; positive edge-trigger; 3-state

SSOP-20L



12. Tape and Reel Information

12.1. Carrier tape dimensions

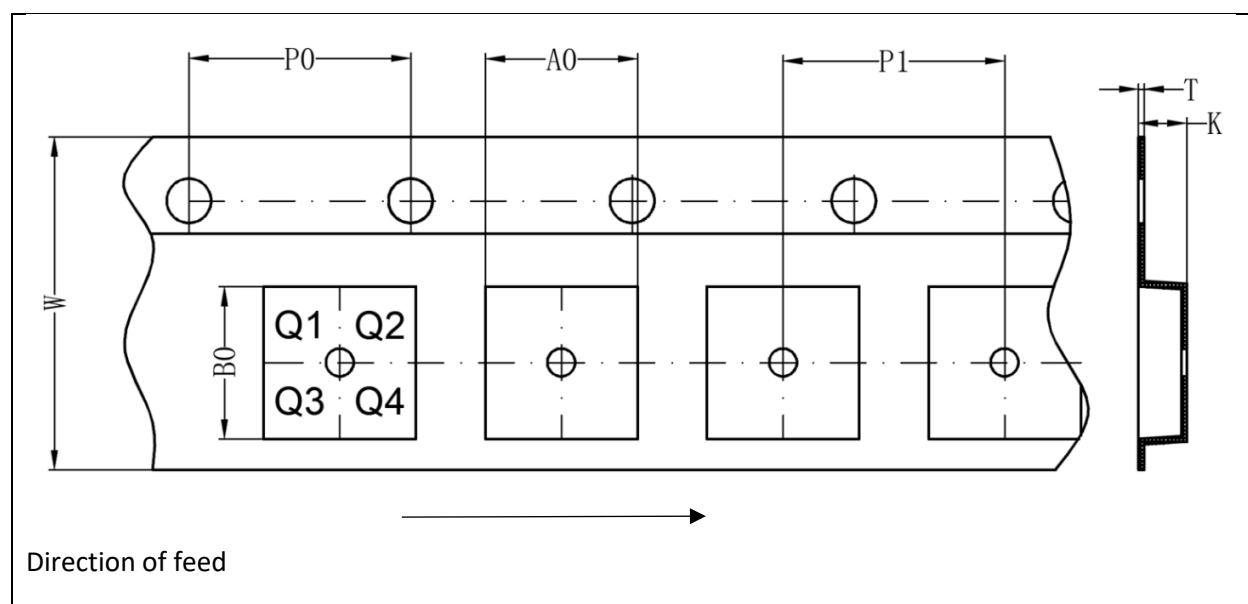


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-20L	10.65	13.2	3.2	0.3	12	24	4	Q1
TSSOP-20L	6.85	6.85	1.7	0.22	8	12	4	Q1
SSOP-20L	8.4	7.75	2.5	0.3	12	16	4	Q1

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12.2.Reel and box dimensions

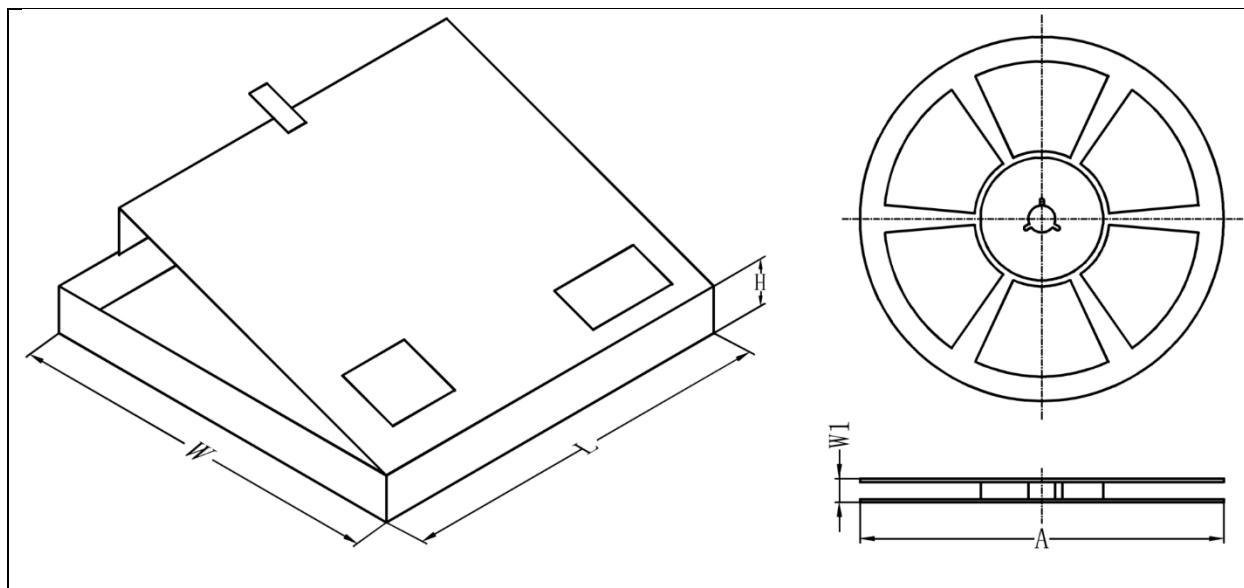


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-20L	D	330	30.4	2000	1	358x340x50
TSSOP-20L	PW	330	18.4	3000	1	358x340x50
SSOP-20L	DB	330	22.4	2000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

EM74HC574; EM74HCT574**Octal D-type flip-flop; positive edge-trigger; 3-state**

13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT574 Rev. 1.0	May 20, 2025	Product datasheet		